MYEONGWON CHAE

Champaign, IL, USA

Education

University of Illinois, Urbana-Champaign, IL, USA — Aug. 2023 – May. 2029

- *Ph.D.,* GPA: 3.96/4.0
- *Master of Science in Architectural Studies,* GPA: 4.0/4.0

Ajou University, Suwon, Korea — Mar. 2014 – Feb. 2022

- *Bachelor of Architecture* (Intensive major course), GPA: 3.7/4.5

Professional Experience

Research Assistant, The Responsive Architecture Lab, University of Illinois, Urbana-Champaign — Dec. 2023 – May. 2026

- Applied machine learning and retrieval-augmented **large language models** to analyze energy consumption patterns and generate structured insights from technical documents for building retrofit decision support.
- Developed data-driven methods to evaluate building energy performance using smart thermostat data, including analysis of HVAC system sizing suitability and operational efficiency.
- Built and calibrated building energy simulation models using **DesignBuilder** and **EnergyPlus** to support performance evaluation and potential energy savings analysis.

Retro-Commissioning team, F&S team, University of Illinois, Urbana-Champaign — May. 2024 – Aug. 2024

- Analyzed energy consumption trends in on-campus buildings and handling HVAC systems plan
- Modeled and calibrated building energy for buildings on campus using **eQuest**

Teaching Assistant, School of Architecture, University of Illinois, Urbana-Champaign — Aug. 2023 – Dec. 2024

- Instructing 12 graduate students on the use of **DesignBuilder** for building energy simulation
- Taught 15 undergraduate students and taking charge as the supplemental instructor for weekly design sessions

Researcher, Urban & Building Energy (UBE) Lab, Ajou University, Suwon, Korea — Feb. 2023 – May. 2023

- Derived new regional heating and cooling start temperature by using building energy usage data to replace the existing balance point temperature

Undergraduate Researcher, Urban & Building Energy (UBE) Lab, Ajou University — May. 2020 – Aug. 2021

- Researched to disclose building energy performance information disclosure and expanded and revitalized utilization system
- Proved energy efficiency diagnosis and remodeling technology for existing public buildings

Honorary Discharge as a Sergeant**,** Republic of Korea Army, Inje-gun, Gangwon, Korea — Mar. 2016 – Dec. 2017

- Completed the Compulsory military service as an administrator and field engineer

Publication

- **Myeongwon Chae**, Yun Kyu Yi. "Retrieval-Augmented Large Language Models for Building Retrofit Recommendations." *18th International IBPSA Building Simulation Conference (BS2027),* Vienna, Austria, August 29 – September 1, 2027 (Submitted).
- **Myeongwon Chae,** You-Joung Kim, Yun Kyu Yi "Data-Driven Assessment of Air-Conditioning System Sizing Using Runtime and Indoor Humidity Performance." *12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings*, Los Angeles, California, USA, May 2026
- **Chae, MyeongWon**, Kim, Hye-Gi, and S.S.Kim "Derivation of heating and cooling start temperature by region using building energy use data." *Journal of the Architectural Institute of Korea, 404 - 405,* Seoul, Korea, April 2023
- H.G.Kim, H.J.Kim, C.H.Jeon, **M.W. Chae**, Y.H.Cho and S.S.Kim "Analysis of Energy Saving Effect and Cost Efficiency of ECMs to Upgrade the Building Energy Code." *Energies 2020, 13(18), 4955*, Seoul, Korea, 21 September 2020

Awards and Honors

- **Fellowship**, Clarence T Paul Fellowship, Illinois, USA — Aug. 2025
- **Scholarship**, JAQ'D Architecture Scholarship, Illinois, USA — May. 2024
- **Fellowship**, Munju Scholarship Foundation, Seoul, Korea — Feb. 2021
- **Dean's List**, Ajou University, Suwon, Korea — Feb. 2021
- **Grand Prize Award**, An Idea Contest to Revitalize the Space of Smart Green Student Plaza, Ajou University, Suwon, Korea — Oct. 2020
- **Encouragement Award**, 2019 Central Urban Regeneration Candidates University Student Architecture Urban Studio, Korean Society of Urban Planning, Seoul, Korea — Nov. 2019

Extracurricular Activities

Korean Student Association (KSA), University of Illinois, Urbana-Champaign, IL, USA — Dec. 2023 – Dec. 2024

- Supported corporate recruiting events, attracted corporate sponsors, and organized academic exchanges between universities in the Midwest.

Student Ambassador (GAIA), Ajou University, Suwon, Korea — May. 2014 – Oct. 2015

- Carried out briefing sessions for admissions procedure and campus tour

Skills

- Python, EnergyPlus, Design Builder, Open Studio, eQuest, Sketch up, Rhino, Auto CAD